

Worksheet

Section 1 – Objective

Q1. A wire carrying a current of 5A is placed perpendicular to a magnetic induction of 2T. The force on each centimeter of the wire is:

- a) 1N
- b) 100N
- c) 0.1N
- d) 10N

Q2. If a soft iron piece is buried under the surface of earth in the north and south direction, then

- a) It will acquire the properties of a magnet
- b) Its properties will not change
- c) It will behave like an insulator
- d) Can't say with surety

Q3. Force acting on a stationary charge Q in the magnetic field B is –

- a) $B Q V$
- b) BV/Q
- c) ZERO
- d) BQ/V

Q4. A proton is moving with velocity 104m/s parallel to the magnetic field of intensity 5Tesla. The force on the proton is –

- a) $8 \times 10^{-15} \text{ N}$
- b) 104 N
- c) $1.6 \times 10^{-19} \text{ N}$
- d) Zero

Q5. A wire of length 'l' is placed in a magnetic field 'B'. If the current in the wire is 'I', then maximum magnetic force on the wire is:

- a) BIL
- b) B/IL
- c) IL/B

d) I/BL

Q6. Which of the following statement is correct?

- a) A magnet may exert force on a charged particle
- b) A charged particle may move undeflected in a magnetic field
- c) Moving charges are the source of magnetic field
- d) All are correct

Q7. A magnetic field may:

- a) Change the velocity of a charged particle
- b) Change the speed of a charged particle
- c) Change the KE of a charged particle
- d) Stop a moving charged particle

Q8. Lenz's law is a consequence of the law of conservation of:

- a) Energy
- b) Momentum
- c) Angular momentum
- d) Charge and mass

Q9. The induced emf produced when a magnet is inserted into a coil does not depend upon:

- a) The number of turns of coil
- b) The resistance of coil
- c) The magnetic moment of the magnet
- d) The speed of approach of the magnet

Q10. Lenz's law

- a) Is the same as the right hand palm rule
- b) Determines the magnitude of an induced emf
- c) Bears no relation to the law of conservation of energy
- d) Is useful in deciding about the direction of an induced emf

Q11. A.C. used in our domestic consumption has a frequency

- a) 60Hz
- b) 50Hz
- c) 30Hz
- d) 100Hz

Q12. A solenoid having an iron core has its terminals connected across an ideal D.C source. If the iron core is removed the current flowing through solenoid is:

- a) Increases
- b) Decreases
- c) Remains unchanged
- d) NOT

Q13. Direction of force acting on a current carrying conductor kept in a magnetic field is given by

- a) Fleming's right hand rule
- b) Fleming's left hand rule
- c) Lens's rule
- d) Faraday's rule

Q14. The unit of induced emf is:

- a) Ampere
- b) Volt
- c) Joule
- d) Electron volt

Q15. 2000 J of energy is needed to heat 1kg of paraffin through 1°C. How much energy is needed to heat 10kg of paraffin through 2°C?

- 1. 4000 J
- 2. 10,000 J
- 3. 20,000 J
- 4. 40,000 J

Q16. The most commonly used thermometric liquid is:

- 1. Water
- 2. Alcohol
- 3. Mercury
- 4. None of these

Q17. Absolute zero corresponds to:

- 1. -273 K
- 2. -273°C
- 3. 273°F
- 4. None of these

Q18 A temperature difference of 30°C in the Fahrenheit scale is:

1. 54°F
2. 27°F
3. 86°F
4. -54°F

Q19. With increase in temperature the density of a substance in general:

1. Increases
2. Decreases
3. First increases then decreases
4. First decreases then increases

Q20. γ_r is coefficient of real expansion of a liquid, γ_a is coefficient of apparent expansion of a liquid and γ_g is coefficient of cubical expansion of the container then the correct relation is:

1. $\gamma_r = \gamma_a \times \gamma_g$
2. $\gamma_r = \gamma_a \div \gamma_g$
3. $\gamma_r = \gamma_a - \gamma_g$
4. $\gamma_r = \gamma_a + \gamma_g$

Q21. Per °C is unit of:

1. α
2. β
3. γ
4. All of these

Q22. Clock's pendulum is made of invar because:

1. It is light in weight
2. It is easily available
3. Its coefficient of linear expansion is low
4. It is cheaply available

Q23. 1 kg of water at 20°C is, mixed with 800g of water at 80°C. Assuming that no heat is lost to the surroundings calculate the final temperature of the mixture:

1. 24.44°C
2. 46.67°C
3. 44.44°C
4. 54.44°C

Q24. How much heat would be required to convert 14kg of ice at 0°C into water of 0°C?

1. 4704 J
2. 4704 kJ
3. 336 J
4. 336 kJ

Q25. The specific heat capacity of water is 4200 J/kg °C. Calculate the heat capacity of 10kg of water per °C

1. 42 J
2. 420 J
3. 42 kJ
4. 420 kJ

Q26. The temperature of a metal coin is increased by 100°C and its diameter increases by 0.15%. Its area increases by nearly:

1. 0.15%
2. 0.60%
3. 0.30%
4. 0.0225%

Q27. The change from solid to vapour directly at a constant temperature is called:

1. Condensation
2. Regelation
3. Vaporization
4. Sublimation

Q28. The specific latent heat of fusion of water is:

1. 80 cal g⁻¹
2. 2260 J g⁻¹
3. 80 J g⁻¹
4. 336 J kg⁻¹

Q29. Expansion in a substance is:

1. Directly proportional to rise in the temperature
2. Inversely proportional to rise in the temperature
3. Independent of rise temperature
4. Cannot say

Q30. The minimum amount of energy required to emit electrons from a metal surface is called the:

- a) Ionization energy
- b) Dissociation energy
- c) Cohesive energy
- d) Work function

Q31. The work function of tungsten coated with barium oxide is nearly:

- a) 5eV
- b) 10eV
- c) 1eV
- d) 0.01eV

Q32. The electron emitter must have:

- a) Low work function and low melting point
- b) High work function and high melting point
- c) High work function and low melting point
- d) Low work function and high melting point

Q33. A TV tube requires an accelerating voltage of the order:

- a) 150 volt
- b) 1000 volt
- c) 20 kV
- d) 500 kV

Q34. Electrons was discovered by:

- a) Neil Bohr
- b) J.J. Thomson
- c) R.A. Millikan
- d) Rutherford

Q35. The charge on an atom is:

- a) Positive
- b) Negative
- c) Neutral
- d) None of these

Q36. Isotone of an element has –

- a) Same number of electrons
- b) Same number of protons
- c) Same number of neutrons
- d) Same number of neutrons and protons

Q37. An isotones of $^{32}_{\text{Ge}}76$ is

- a) $^{32}_{\text{Ge}}77$
- b) $^{33}_{\text{Ge}}77$
- c) $^{36}_{\text{Ge}}77$
- d) $^{34}_{\text{Ge}}77$

Q38. The triad of nuclei that is isotonic is:

- a) $^{6}_{\text{C}}14, ^{7}_{\text{N}}15, ^{9}_{\text{F}}17$
- b) $^{6}_{\text{C}}12, ^{7}_{\text{N}}14, ^{9}_{\text{F}}19$
- c) $^{6}_{\text{C}}14, ^{7}_{\text{N}}14, ^{9}_{\text{F}}17$
- d) $^{6}_{\text{C}}14, ^{7}_{\text{N}}14, ^{9}_{\text{F}}19$

Q39. Which particle is used to bombard $^{13}_{\text{Al}}28$ to give $^{15}_{\text{P}}31$ and a neutron?

- a) α -particle
- b) Deuteron
- c) Proton
- d) Neutron

Q40. Gamma rays are:

- a) High energy electrons
- b) Low energy electrons
- c) High energy electromagnetic waves
- d) High energy positrons

Q41. Which of the following radiation is least penetrating?

- a) α -particles
- b) β^- particles
- c) X-rays
- d) γ – radiations

Q42. Which of the following radiation suffers maximum deflection in a magnetic field ?

- a) α -particles
- b) β^- -particles

- c) X-rays
- d) Γ – radiations

Section 2 – Subjective

Q1. How can we find the direction of magnetic force on current carrying conductor?

Q2. What is an electromagnet? How does it differ from a permanent magnet? State three factors on which strength of an electromagnet depends ?

Q3. State Fleming's left hand rule.

Q4. State the principle of a motor.

Q5. Write faraday's laws of electromagnetic induction.

Q6. Write the five differences between AC and DC.

Q7. Give the working of slip rings in the AC generator.

Q8. State the types of transformer available.

Q9. Define the following

- a) Melting point
- b) Latent heat
- c) Vaporization

Q10. Distinguish between

- a) Heat and temperature
- b) Heat capacity and specific heat capacity

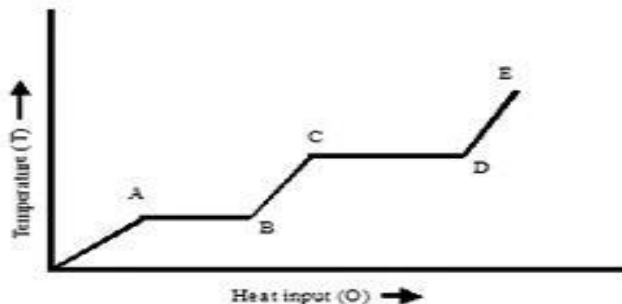
Q11. Calculate the amount of heat required to convert 1 kg of ice at -10°C into steam at 100°C at normal pressure. Specific heat capacity of ice = $2100 \text{ J kg}^{-1} \text{ K}^{-1}$. Latent heat of fusion of ice = $3.36 \times 10^6 \text{ J kg}^{-1}$, specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$ and latent heat of vaporization of water = $2.25 \times 10^6 \text{ J kg}^{-1}$.

Q12. An iron piece of mass 2 kg has a thermal capacity of $966 \text{ J/}^{\circ}\text{C}$.

How much heat is needed to warm it by 15°C ?

What is its specific heat capacity in SI units?

Q13. A source of heat supplies heat at a constant rate to a solid cube. The variation of the temperature of the cube with the heat supplied is shown in the figure given below.



What does the slope of the part DE represent?

If $CD = 2.5 AB$, what does it mean?

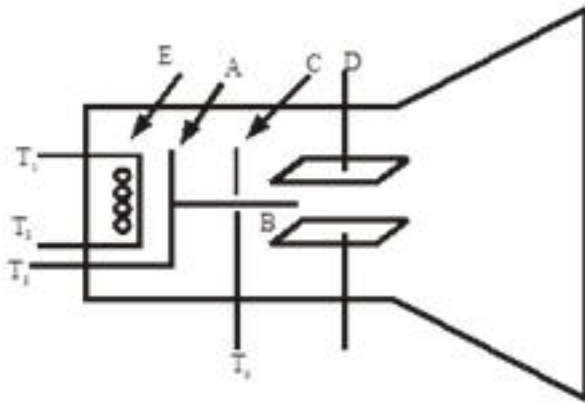
Q14. Define the term work function of a metal

Q15. State any three properties of α , β and γ rays.

Q16. What do you mean by thermionic emission? State its two properties

Q17. $X\text{RnY} \rightarrow {}^{86}\text{Po}^{218} + \alpha\text{-particles}$. Find X And Y.

Q18. In the labelled diagram answer the following



What is represented by symbols A, B, C, D and E?

States the purpose of A, B, C and D.

Q19. Give one use of radio – isotopes. What is meant by background radiations?

Q.20 .Name a common device where a hot cathode ray tube is used.

State the two properties which a substance should possess when used as thermionic emitter.